Proofs and Mathematical Reasoning
Strong Induction Batch

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November 6, 2026

Contents

1 Number of Subsets **2**

2 Binary Representation **2**

3 Prime Factorisation **3**

4 Handshake Lemma **4**

About

Worked proofs from Rosen, *Discrete Mathematics and Its Applications*, ch.4-10. Each section records the question, my first attempts, where they went wrong, what made it click, and the clean proof. The wrong turns are the point.

1 Number of Subsets

Question. Prove by induction that a set with n elements has exactly 2^n subsets.

First I checked small cases because I always feel better when I see the numbers work. For $n = 0$, the only set is $\varnothing$ and its only subset is $\varnothing$, so count $= 1 = 2^0$. For $n = 1$, set $\{a\}$, subsets $\varnothing$ and $\{a\}$, count $= 2 = 2^1$. For $n = 2$, set $\{a, b\}$, subsets $\varnothing$, $\{a\}$, $\{b\}$, $\{a, b\}$, count $= 4 = 2^2$. For $n = 3$, count $= 8 = 2^3$. Pattern obvious.

The inductive step felt less obvious. Let $S = \{a_1, \ldots, a_k, x\}$ with x the new element. The idea I eventually landed on: every subset of S either contains x or does not, so split into two piles. The pile that does not contain x is exactly $\mathcal{P}(\{a_1, \ldots, a_k\})$, which by the inductive hypothesis has 2^k elements. The pile that does contain x is everything of the form $T \cup \{x\}$ where $T \subseteq \{a_1, \ldots, a_k\}$, so the number of such subsets equals the number of choices of T, again 2^k. Total $2^k + 2^k = 2^{k+1}$.

The mistakes. First attempt was to count subsets of size r and sum: $|\mathcal{P}(S)| = \sum_{r=0}^{n} \binom{n}{r}$, then invoke the binomial theorem to get 2^n. That works but it outsources the hard bit; the binomial theorem is itself proved by induction, so it feels like cheating. Second attempt was to argue "each element independently has 2 choices", which is correct intuition but formalising it without assuming the answer needs a bijection to $\{0, 1\}^{k+1}$, basically as hard as the original problem. Third attempt was to write $\mathcal{P}(S) = \mathcal{P}(T) \cup \{U \cup \{x\} : U \in \mathcal{P}(T)\}$ and then panic about whether this union is disjoint. It is trivially disjoint because no subset can both contain x and not contain x. I was overthinking.

Proof. Claim: if $|S| = n$, then $|\mathcal{P}(S)| = 2^n$.

Base case $n = 0$: the only zero-element set is $\varnothing$, whose only subset is $\varnothing$, so $|\mathcal{P}(\varnothing)| = 1 = 2^0$.

Inductive hypothesis: assume every k-element set has exactly 2^k subsets.

Inductive step: let S have $|S| = k + 1$. Fix $x \in S$ and let $T = S \setminus \{x\}$, so $|T| = k$. Every subset of S either contains x or does not:

$$\mathcal{P}(S) = \{A \subseteq S : x \notin A\} \cup \{A \subseteq S : x \in A\}.$$

This union is disjoint. The first part is $\mathcal{P}(T)$, so by the inductive hypothesis it has 2^k elements. For the second part, the map $B \mapsto B \cup \{x\}$ is a bijection from $\mathcal{P}(T)$ to $\{A \subseteq S : x \in A\}$, so the second part also has 2^k elements. Therefore

$$|\mathcal{P}(S)| = 2^k + 2^k = 2 \cdot 2^k = 2^{k+1}.$$

By induction, every n-element set has 2^n subsets. □

The key idea to remember: adding one element doubles the number of subsets, because every old subset spawns two new ones (itself, and itself plus the new element). That doubling is the whole engine of the proof.

2 Binary Representation

Question. Prove by strong induction that every positive integer can be written as a sum of distinct powers of 2.

First instinct was weak induction because that is what I am comfortable with. I listed small cases: $3 = 2 + 1$, $4 = 4$, $5 = 4 + 1$, $6 = 4 + 2$, $7 = 4 + 2 + 1$, $8 = 8$. The pattern exists but

I could not see a general rule for going from k to $k+1$. Sometimes $k+1$ is a single power, sometimes a power plus something. Adding 1 can carry. For instance $7 = 4+2+1$; add 1 and I have $4+2+1+1$, repeated 1, combine to $4+2+2$, combine to $4+4$, then 8. Binary addition changes multiple powers at once, and I could not formalise it. Dead end.

The mistakes. First, forcing weak induction on the step $n \to n+1$ — the wrong tool, because adding 1 can collide with existing 2^0 and cascade. Second, thinking "binary is literally a sum of distinct powers of 2, so obviously every integer has one" — circular, since binary is exactly what I am trying to prove exists. Third, looking for one closed formula or algorithm for the representation of any n — not induction, and I could not find one without already assuming binary works.

What made it click. Strong induction lets me reduce to any smaller number, not just the one immediately below. So instead of asking "how do I get from k to $k+1$", ask "what smaller number can I reduce $k+1$ to?". The answer is parity. If $k+1$ is even, $k+1 = 2m$ with $m < k+1$; use the representation of m and double every power, which preserves distinctness because $2 \cdot 2^a = 2^{a+1}$. If $k+1$ is odd, $k+1 = 2m+1$; use the representation of m, double every power, and add 2^0. The 2^0 is distinct from the doubled powers because those are all at least 2^1.

Proof. Claim: every positive integer n is a sum of distinct powers of 2.

Base case $n = 1$: $1 = 2^0$.

Inductive hypothesis: assume every integer j with $1 \le j \le k$ has such a representation.

Inductive step: let $n = k+1$.

Case 1, $k+1$ even. Write $k+1 = 2m$, so $m = (k+1)/2$. Then $1 \le m \le k$, and by the inductive hypothesis $m = 2^{a_1} + \cdots + 2^{a_r}$ with distinct a_i. Therefore

$$k+1 = 2m = 2^{a_1+1} + \cdots + 2^{a_r+1},$$

and the exponents $a_i + 1$ are still distinct.

Case 2, $k+1$ odd. Write $k+1 = 2m+1$, so $m = k/2$. Since $k+1 \ge 3$ here (the only positive odd below 3 is 1, handled in the base case), $1 \le m \le k$, and by the inductive hypothesis $m = 2^{a_1} + \cdots + 2^{a_r}$ with distinct a_i. Therefore

$$k+1 = 2m+1 = 2^{a_1+1} + \cdots + 2^{a_r+1} + 2^0,$$

and each $a_i + 1 \ge 1$, so no exponent equals 0, so 2^0 is distinct from all the doubled powers.

Both cases work, so by strong induction every positive integer has such a representation. □

The lesson to keep: when weak induction gets stuck on $k \to k+1$, that is the signal to reach for strong induction and ask "what smaller number can I reduce to?". Here the reduction is parity — halve and double, or halve, double, add 1. The trap that almost caught me was thinking binary was the answer. Binary is what the theorem says exists; using it in the proof assumes the conclusion.

3 Prime Factorisation

Question. Prove by strong induction that every integer $n \ge 2$ can be written as a product of primes. This is the existence half of the Fundamental Theorem of Arithmetic.

Small cases first: $2 = 2$, $3 = 3$, $4 = 2 \cdot 2$, $5 = 5$, $6 = 2 \cdot 3$, $7 = 7$, $8 = 2^3$, $9 = 3^2$, $10 = 2 \cdot 5$, $11 = 11$, $12 = 2^2 \cdot 3$. All good. So assume every integer from 2 up to k has a factorisation, and show $k+1$ does. If $k+1$ is prime, then $k+1 = k+1$ is a single prime. If $k+1$ is composite, then $k+1 = ab$ with $1 < a, b < k+1$, both therefore between 2 and k. By the strong inductive hypothesis $a = p_1 \cdots p_r$ and $b = q_1 \cdots q_s$ with each p_i, q_j prime, so $k+1 = p_1 \cdots p_r q_1 \cdots q_s$, product of primes.

The mistakes. First attempt was weak induction: assume k has a factorisation, prove $k+1$ does. This gets stuck immediately on composite $k+1 = ab$ where a and b are much smaller

than k. For $k+1=6$, $a=2$, $b=3$, but the weak inductive hypothesis only tells me about 5. Dead end. The key difference: weak gives one fact, strong gives every fact from 2 up to k. Second attempt tried to avoid induction entirely by "keep dividing by primes until you cannot" — that is correct in spirit but formalising termination is a mess, and induction is the clean way to formalise it. Third attempt got tangled up in what "prime" and "composite" actually mean; I had to work out that composite means a divisor d with $1 < d < n$, which gives a factorisation $n = d \cdot (n/d)$ with both factors strictly between 1 and n. I had to establish that, not just assert it. Fourth attempt worried about the two factorisations sharing primes — not a problem, because I am proving existence, not uniqueness. $12 = 2 \cdot 2 \cdot 3$, the 2 appears twice, fine.

Proof. Claim: every integer $n \geq 2$ is a product of primes.

Base case $n = 2$: 2 is prime, so $2 = 2$ is a product of one prime.

Inductive hypothesis: assume every integer m with $2 \leq m \leq k$ is a product of primes.

Inductive step: let $n = k + 1$.

Case 1, $k+1$ prime. Then $k+1 = k+1$ is a product of primes.

Case 2, $k+1$ composite. By definition there exists a divisor d of $k+1$ with $1 < d < k+1$. Let $m = (k+1)/d$. Then m is an integer and $1 < m < k+1$, since $d \cdot m = k+1$ and $1 < d < k+1$ force $1 < m < k+1$. So $k+1 = d \cdot m$ with $2 \leq d \leq k$ and $2 \leq m \leq k$. By the strong inductive hypothesis, $d = p_1 \cdots p_r$ and $m = q_1 \cdots q_s$ with each p_i, q_j prime. Therefore $k+1 = p_1 \cdots p_r \cdot q_1 \cdots q_s$, a product of primes.

Both cases work, so by strong induction every integer $n \geq 2$ has a prime factorisation. □

What I am not proving here: uniqueness. The Fundamental Theorem has two halves. This proof gives existence. Uniqueness — no two different multisets of primes multiply to the same number — is a separate and harder proof, usually by contradiction using Euclid's lemma ($p \mid ab$ implies $p \mid a$ or $p \mid b$). Existence is what the question asked for.

4 Handshake Lemma

Question. Prove by induction that in any simple undirected graph, the sum of all vertex degrees equals twice the number of edges: $\sum_{v \in V} \deg(v) = 2|E|$.

I had never seen a graph in this sense before. When someone said "graph" I thought of plotting $y = f(x)$ on axes. So the first thing I had to do was figure out what this question was even about, and the honest answer is that I sat with the definitions for a while before anything made sense.

The definition I found in Rosen: a *simple undirected graph* is a pair $G = (V, E)$ where V is a set of vertices and E is a set of unordered pairs $\{u, v\}$ of distinct vertices. No loops (no edge from a vertex to itself) and no multiple edges (at most one edge between any two vertices). The word "undirected" means the edge $\{u, v\}$ is the same as $\{v, u\}$; there is no arrow, no direction.

The *degree* of a vertex v, written $\deg(v)$, is the number of edges incident to v. Equivalently, the number of other vertices u such that $\{v, u\} \in E$.

That definition took me a solid ten minutes to actually internalise. The notation $\{u, v\} \in E$ looks like set membership but it means "there is an edge between u and v". Once I started reading it as "u and v are joined", it stopped being scary.

I then drew examples. Not just on paper — I actually drew them, and counted degrees by hand. A graph with one vertex and no edges: $\deg(a) = 0$, sum $= 0$, and $2|E| = 0$. Good. A graph with two vertices and one edge between them: $\deg(a) = \deg(b) = 1$, sum $= 2$, and $2|E| = 2 \cdot 1 = 2$. Good. A path of three vertices $a - b - c$: $\deg(a) = 1$, $\deg(b) = 2$, $\deg(c) = 1$, sum $= 4$, and $2|E| = 2 \cdot 2 = 4$. Good. A triangle: each vertex has degree 2, sum $= 6$, and $2|E| = 2 \cdot 3 = 6$. Good. A graph with five vertices and no edges: all degrees 0, sum $= 0$, and $2|E| = 0$. Good.

Seeing the numbers work was the first real progress. But "the formula is true for these examples" is not a proof, and I was aware of that.

Then the real question: what do I induce on? There are two obvious candidates — the number of vertices, or the number of edges. I tried vertices first because that is what came to mind. Assume true for every graph with k vertices, prove for every graph with $k+1$ vertices. To do that I would need to build a $(k+1)$-vertex graph from a k-vertex graph, which means adding one vertex and some edges from it to existing vertices. If the new vertex is adjacent to t existing vertices, then it contributes t to the sum of degrees (its own degree is t), and the t neighbours each gain 1 to their degree, so the total sum goes up by $2t$. That is exactly $2\times$ (number of new edges), so if the old sum was $2|E_{\text{old}}|$, the new sum is $2|E_{\text{old}}| + 2t = 2|E_{\text{new}}|$. The algebra works.

The problem was that I could not cleanly state "every $(k+1)$-vertex graph arises by adding one vertex to some k-vertex graph". Technically yes — remove any vertex from a $(k+1)$-vertex graph and you get a k-vertex graph — but the "any vertex" is doing a lot of quiet work, because the number of edges t to the removed vertex depends on which vertex I pick, and the induction hypothesis has to apply to the specific resulting graph. I could not get the bookkeeping airtight, and I knew that was a warning sign.

So I tried edges instead. Assume true for every graph with k edges, prove for every graph with $k+1$ edges. To get from a k-edge graph to a $(k+1)$-edge graph, add one edge. To get from a $(k+1)$-edge graph to a k-edge graph, remove one edge. Removing an edge is much cleaner than removing a vertex, because the only thing that changes is the two endpoints. In a simple graph, an edge is $\{u, v\}$ with $u \neq v$, so removing it drops $\deg(u)$ by 1, drops $\deg(v)$ by 1, and leaves every other vertex alone. Two degree counts drop by exactly 1 each. That is it. No bookkeeping about how many neighbours the removed vertex had, no case analysis.

That is when it clicked. Induction on the number of edges is the right choice because edge removal has a clean, local effect.

Proof. Claim: for any simple undirected graph $G = (V, E)$,

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Proof by induction on $k = |E|$.

Base case $k = 0$: G has no edges, so every vertex has degree 0, and $\sum_{v \in V} \deg(v) = 0 = 2 \cdot 0 = 2|E|$.

Inductive hypothesis: assume every simple undirected graph with exactly k edges satisfies the claim.

Inductive step: let $G = (V, E)$ be a simple undirected graph with $|E| = k+1$. Since $k+1 \geq 1$, the edge set is nonempty, so pick any edge $e = \{u, v\} \in E$. Because G is simple, $u \neq v$. Define $G' = (V, E \setminus \{e\})$ by removing e. Then G' is a simple undirected graph with exactly k edges, so by the inductive hypothesis

$$\sum_{w \in V} \deg_{G'}(w) = 2k.$$

The only edge that differs between G and G' is e, and e is incident to u and v (and to no other vertices). Therefore

$$\deg_G(u) = \deg_{G'}(u) + 1, \quad \deg_G(v) = \deg_{G'}(v) + 1,$$

and $\deg_G(w) = \deg_{G'}(w)$ for all $w \in V \setminus \{u, v\}$. Because $u \neq v$, these are two distinct increments, not the same one applied twice. Summing:

$$\sum_{w \in V} \deg_G(w) = \sum_{w \in V} \deg_{G'}(w) + 1 + 1 = 2k + 2 = 2(k+1) = 2|E|.$$

By induction, every simple undirected graph satisfies the claim. □

The thing I want to remember: induction on edges is cleaner than induction on vertices because removing an edge has a local effect (only two degrees change), whereas removing a vertex drags in its neighbours. Also, the condition $u \neq v$ matters. If loops were allowed, adding or removing a loop would change the degree count depending on convention (some authors count a loop as $+2$ to the degree, some as $+1$), and the lemma would need a caveat. For simple graphs, no loops, everything is clean.

The intuition that holds the whole thing together: every edge has exactly two endpoints, and it is counted exactly once per endpoint in the degree sum. So the total degree count is twice the number of edges, because every edge gets counted twice — once at each end.